

Supporting Information for "GICs in Detailed Irish Power Network Model"

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1. GIC datasets for 5 historic events
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Introduction

The five event databases (August 2015, September 2015, October 2015, December 2015 and March 2016) have measured and calculated GIC values at the 400/220 kV Woodland substation. The "August 2015 GIC Data for Different Grounds" database contains

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measured and calculated GIC data at the Woodland substation for different substation ground values used.

GIC Data Set August 2015 GIC. Measured and calculated GIC data at the Woodland substation for the August 2015 geomagnetic storm.

GIC Data Set September 2015 GIC. Measured and calculated GIC data at the Woodland substation for the September 2015 geomagnetic storm.

GIC Data Set October 2015 GIC. Measured and calculated GIC data at the Woodland substation for the October 2015 geomagnetic storm.

GIC Data Set December 2015 GIC. Measured and calculated GIC data at the Woodland substation for the December 2015 geomagnetic storm.

GIC Data Set March 2016 GIC. Measured and calculated GIC data at the Woodland substation for the March 2016 geomagnetic storm.

GIC Data Set August 2015 GIC for Different Grounds Measured and calculated GIC data at the Woodland substation for the August 2015 geomagnetic storm. Calculated GICs are for different substation ground values used in the power network model.